ABHINAV TRIPATHI

PYTHON & JAVA DEVELOPER

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PROFESSIONAL SUMMARY

Python & Java Developer and MCA scholar at Amity University Online (BCA Graduate, Amity University Lucknow) specializing in application development, robust backend architectures, and multi-agent AI systems. Proven track record engineering award-shortlisted platforms across fintech, healthcare incident remediation, and on-device machine learning with full data privacy compliance.

EDUCATION

Amity University Online — Noida, UP 2026 – 2028
Master of Computer Applications (MCA) | ***In Progress (2026 – 2028 Batch)***

Amity University — Lucknow, UP 2023 – 2026
Bachelor of Computer Applications (BCA) | ***GPA: 7.50 / 10.0***

Relevant Coursework: Data Structures & Algorithms (Ongoing Learning), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Software Engineering.

Central Academy — Lucknow, UP Lucknow, UP
CBSE Class X (Secondary): 79.0% (01/2020) • **CBSE Class XII (Senior Secondary):** 69.0% (01/2022)

TECHNICAL SKILLS

Programming Languages:	**Python** (Core, OOP, Scripting, AsyncIO), **Java** (Core Java, OOP, Collections Framework, Exception Handling)
Data Structures & Algorithms:	**Ongoing Learning** (Arrays, Strings, Two Pointers, Binary Search, Recursion, Time & Space Complexity)
Web & Backend:	FastAPI, RESTful APIs, Asynchronous Programming, HTML5, CSS3, JSON Schema
Databases & Systems:	Relational Databases (SQL, MySQL, SQLite), SQLAlchemy ORM, Linux/Unix Terminal
AI & Developer Tools:	Multi-Agent Systems, LangGraph, Qualcomm AI Hub, ONNX Runtime, Git, GitHub, VS Code
Spoken Languages:	English (Fluent), Hindi (Native), French (Beginner)

TECHNICAL PROJECTS

SAATHI — Autonomous Relationship Intelligence Banking Agent / *Python, FastAPI, LangGraph* GitHub • GFF Shortlist

- Architected a proactive banking agent platform shortlisted in the **SBI Hackathon at Global Fintech Fest 2026** (organized by NPCI & FCC).
- Engineered a **"One Face, 12 Brains"** multi-agent orchestrator in Python using **LangGraph** (`SENTINEL`, `KARMA`, `SHIELD`), cutting UPI grievance turnaround from **30 days to 4.2 minutes** via automated transaction log parsing.
- Designed the `GUARDIAN` deterministic veto engine ensuring 100% compliance with **DPDPA 2023** & **RBI FREE-AI guidelines**, backed by an asynchronous **FastAPI** SSE streaming backend.

MedTrace — Autonomous Clinical Software Remediation Platform / *Python, Multi-Agent Orchestration, REST APIs* Live Demo • GitHub

- Architected an autonomous healthcare incident remediation platform for the **IBM Bob 2.0 Hackathon**, running a 5-stage concurrent multi-agent pipeline to isolate root causes and synthesize AST code patches within **90 seconds**.
- Implemented automated test synthesis and regression suite (**16/16 passed integration tests**) with zero Protected Health Information (PHI) logging, meeting strict **HIPAA §164.312** and **EU AI Act Article 50** standards.

MediSense Edge — On-Device Clinical Decision Support Suite / *Python, Qualcomm AI Hub, ONNX/QNN Runtime, FastAPI* GitHub

- Engineered a 100% on-device clinical intelligence suite for Snapdragon X Elite PCs submitted to the **Qualcomm Snapdragon AI Lab Hackathon**, running completely air-gapped with **0 KB cloud data egress** for total HIPAA/GDPR sovereignty.
- Quantized and deployed 6 models (Phi-3.5 Mini 42ms, Whisper Base 18ms, DenseNet-121, BioBERT) via **Qualcomm AI Hub** onto the **45 TOPS Hexagon NPU**, operating at an ultra-low **1.4W power envelope**.

StaffPulse — Workforce & Attendance Management System / *Web Architecture, Python, SQLite, Automated Payroll* Live Demo • GitHub

- Architected an enterprise attendance platform featuring split-shift tracking (Day 8AM–8PM / Night 8PM–8AM), biometric face-capture verification, and real-time burnout pulse analytics.
- Engineered client-side cryptographic storage (RC4 cipher, SHA-256 session hashing) with zero-server data sovereignty, automated payroll tier computation, and instant payslip PDF generation.

Aora AI — Web3 Behavioral AI Interceptor / *Python Backend, AI Agent Logic, REST APIs* Live Demo • GitHub

- Developed an automated behavioral AI interception prototype to analyze transaction intent and identify anomalous on-chain wallet interactions in real time.

HONORS, CERTIFICATIONS & LEADERSHIP

- **National Shortlist — SBI Hackathon @ Global Fintech Fest 2026:** Shortlisted into the Idea Submission phase by NPCI, FCC, and HackCulture from thousands of nationwide participants for SAATHI AI banking platform.
- **Certificate of Completion — IBM Bob 2.0 Hackathon (Lablab.ai / NativelyAI):** Awarded for outstanding performance, attendance, and submitting MedTrace (Cert ID: CMULAQ6XC00H7S601WX38KT9U).
- **NPTEL Certification in Design Thinking:** Scored **69.0%**, Ministry of Education & IIT (May 2024).
- **AI for Business Professionals:** Professional Certification by HP LIFE (Oct 2025).
- **Volunteer — Climate Conservation (Vartaleap):** Praveera Socio Cultural Foundation, Lucknow (Jan 2024); conducted community sessions to promote climate action and conservation awareness.
- **Problem Solving (Ongoing Learning):** Actively practicing foundational Data Structures & Algorithms (Arrays, Two Pointers, Binary Search, Recursion) in Java & Python.